

MAY ZIN MG

Senior Software Engineer | AI-Enabled Applications, Backend & Cloud | Azure | .NET | SQL

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Microsoft Certified: SQL AI Developer Associate (DP-800) | Azure Solutions Architect Expert | PMP

PROFESSIONAL SUMMARY

Senior Software Engineer specialising in backend, cloud and AI-enabled application engineering, with 15+ years building enterprise systems, APIs, integrations, databases and production automation. Recent hands-on work extends this foundation into agentic AI applications using Python, Microsoft Agent Framework, Azure OpenAI, tool/function calling, structured workflows, testing, guardrails and human-controlled execution. Strong production background in .NET/C#, REST/gRPC APIs, SQL Server, Azure cloud services, CI/CD, Docker and enterprise integration, with experience taking requirements through design, implementation, deployment, troubleshooting and operational support. Microsoft Certified: SQL AI Developer Associate (DP-800), Azure Solutions Architect Expert and PMP certified.

CORE SKILLS

AI Applications: Azure OpenAI, Microsoft Agent Framework, agentic workflows, tool/function calling, prompt engineering, RAG concepts, human-in-the-loop validation, guardrails, execution tracing, pytest

Cloud & Platform: Azure Data Factory, Synapse, Data Lake/Blob Storage, Functions, Logic Apps, Container Apps, Key Vault, Managed Identity, App Services

Python & Automation: Python scripting/data processing, Airflow, dbt, DuckDB, Kafka project exposure, NDJSON/Parquet processing, workflow automation

Backend & APIs: C#, ASP.NET Core, .NET 8, gRPC, REST APIs, microservices, FastAPI exposure, enterprise integration, service design

SQL & Data: SQL Server/T-SQL, stored procedures, query optimisation, PostgreSQL, data modelling, ETL/ELT, ingestion, validation, reconciliation

DevOps & Reliability: Azure DevOps, GitHub Actions, CI/CD, Docker, retry/recovery, observability, secure configuration, automated testing, troubleshooting

PROFESSIONAL EXPERIENCE

AZ Digital Pte Ltd, Singapore | 2015 - Present | 11-year progression

Career progression from enterprise backend and systems engineering into cloud platform delivery, production automation, data integration and AI-ready application foundations across finance, POS, logistics, membership and reporting systems.

Cloud & Platform Engineer | 2020 - Present

- Designed and delivered cloud-hosted backend and integration services using .NET/C#, REST/gRPC APIs, SQL Server and Azure, supporting enterprise finance, logistics, POS, membership and reporting workflows.
- Built API- and event-driven integrations across enterprise applications and cloud services, translating business requirements into secure, maintainable system interfaces and reliable data flows.
- Automated recurring operational and reporting workflows using Azure Functions, Logic Apps, Hangfire and scripting, reducing selected manual support effort by an estimated 30-40%.
- Improved selected high-volume reporting and application queries by an estimated 30-40% through SQL Server stored procedure and query optimisation.
- Containerised and deployed cloud-native services using Docker, Azure Container Apps, Key Vault, Managed Identity and CI/CD, while applying validation, retry/recovery, monitoring and production troubleshooting practices.

Backend & Systems Engineer | 2015 - 2020

- Built and maintained .NET backend services, REST APIs and enterprise integrations connecting finance, logistics, POS, membership and reporting systems.
- Designed SQL-backed operational workflows and data interfaces supporting transactional and reporting processes across multiple business domains.
- Owned core backend components and production support, investigating failures across application, database and integration layers and implementing sustainable fixes.

Earlier roles: CalendarOne, Singapore (2012-2015, 2008-2011) | National Library Board, Singapore (2011-2012) | Theseus Computer Help Centre, Myanmar (2004-2006)

SELECTED AI, BACKEND & DATA PROJECTS

Talos Agent Lab - Agentic AI Applications | Python, Microsoft Agent Framework, Foundry Local, Azure OpenAI, pytest
Built and tested a permission-gated tool-calling agent with execution tracing, controlled tool access and automated behavioural tests. Investigated hallucinated tool/data cases, refined instructions and guardrails, and compared grounded behaviour across local and cloud models, focusing on reliability, observability and safe integration of agents with external systems.

Linear B - Agentic Data Platform (In Progress) | Python, DuckDB, Airflow, dbt, LLM Integration
Developing a medallion-style research data platform with deterministic ingestion and transformation layers feeding an LLM-assisted interpretation workflow, separating trusted data preparation from AI reasoning and human review.

AutoBill Invoice Engine | C#, .NET 8, gRPC, QuestPDF, Azure Blob Storage, Docker
Built a cloud-native gRPC service with modular API, document-generation and storage components, demonstrating production backend architecture, secure cloud integration and deployable service design.

IoT Wind Farm - Edge to Insights | Python, Kafka, Redis, Airflow, DuckDB, Parquet, Docker, GitHub Actions
Built an end-to-end streaming analytics pipeline where simulated turbine telemetry flows through Kafka, Redis-based deduplication, Bronze NDJSON, and Airflow/DuckDB transformations into Silver Parquet and Gold KPI outputs.

ProMan DataPlat ELT Pipeline | Python, Airflow, dbt, DuckDB, Azurite, Parquet, Power BI
Built an Azure-style ELT pipeline using raw blob-style storage, Airflow orchestration, dbt/DuckDB bronze-silver-gold models, data-quality checks and BI-ready outputs, demonstrating reusable data integration and modelling patterns.

EDUCATION & CERTIFICATIONS

MSc Business Intelligence (In Progress) IU International University of Applied Sciences, Germany 2024 - Present	• Microsoft Certified: SQL AI Developer Associate (DP-800) • Azure Solutions Architect Expert
MSc Information Systems Nanyang Technological University (NTU), Singapore 2012 - 2014	• PMP • Google Advanced Data Analytics
BSc Information Systems & Software Engineering Oxford Brookes University 2006 - 2007	• Oracle Cloud Infrastructure Data Science Professional

Portfolio: github.com/mayzinmg